Assignment 4: Weighted Campus Route Finder (GBFS and A*)

Group 10B

Nomwesigwa Keith	23/U/16071/EVE	Musheija Abraham	23/U/12139/EVE
Wambui Mariam	23/U/18494/PS	Naluyange Kevin	18/U/23394/EVE
Lukaanya George	23/U/0696	Nabirye Anitah	23/U/12826/EVE

1. Problem formulation (Part A)

We kept the five problem components from Assignment 3: initial state = the source place; actions = walk to any directly connected place (alphabetical order); transition model = the action "walk to X" leads to state X; goal test = state equals the destination; step/path cost = edge weight in metres, summed along the route. The new thing is that every place now has a map position, so we can estimate the metres still to go.

A1. Evaluation functions. UCS: f(n) = g(n), uses only the cost already paid. GBFS: f(n) = h(n), uses only the estimated cost to go. A*: f(n) = g(n) + h(n), uses both, i.e. the estimated total cost of the best route through n.

A2. Admissible. A heuristic is admissible if it never overestimates the true cost of the cheapest route from n to the goal. Straight-line distance is admissible here because every edge weight is at least the straight-line distance between its two ends (verified in B1). A route's cost is the sum of its weights, which is ≥ the sum of its straight segments, which by the triangle inequality is ≥ the single straight line. Rounding down can only lower h further.

A3. Consistent. h is consistent if $h(u) \le c(u,v) + h(v)$ for every edge $u \to v$. Write d for straight-line distance: $d(u,G) \le d(u,v) + d(v,G)$ (triangle inequality) and $d(u,v) \le w(u,v)$ (weight property), so $d(u,G) \le w + d(v,G)$. Because w is a whole number, taking the floor of both sides keeps the inequality: $\lfloor d(u,G) \rfloor \le w + \lfloor d(v,G) \rfloor$. With a reached table, consistency guarantees that the first time A* removes a state from the frontier its g is already the cheapest possible, so A* never needs to re-open a state and the answer is optimal.

A4. By hand. Education (665, 570) to Veterinary Medicine (1310, 1275): $\Delta x = 645$, $\Delta y = 705$, $h = \text{isqrt}(645^2 + 705^2) = \text{isqrt}(913\,050) = 955$ m. The optimal UCS route (Problem 2) costs 1140 m, so the gap is 185 m: h underestimates by about 16 %, which is allowed for an admissible heuristic.

A5. The goal changes h. h(place, goal) measures the distance to a particular goal. For example h(School of Law, Law) = 0 but h(School of Law, Veterinary Medicine) = 1184. A single table with one value per place would be right for one destination only; the program must compute h afresh for each query (and the scan treats A→B and B→A as different problems).

A6. Predict. Yes, GBFS can return a route that is not the cheapest, because it ignores g(n): it follows whatever looks closest to the goal, even if reaching it was already very expensive.

2. Design: what the program does, step by step

2.1 What changed since Assignment 3

- **Data:** `COORDINATES`, `H_SCALE_PERCENT` and h(place, goal) = ⌊straight-line distance⌋ × scale ÷ 100 using `math.isqrt` so every machine gets identical integers. The graph is a weighted adjacency dictionary {place: {neighbour: weight}} stored in alphabetical order.
- **Key functions** (the only thing that distinguishes the priority-queue strategies): `ucs_key` = g, `gbfs_key` = h, `astar_key` = g + h; each takes (node, goal). Heap entries are (key, state_name, counter, node) → smallest key, then alphabetically first name, then earliest inserted.
- **Reached rule** (a flag on each Strategy): BFS, DFS and GBFS add a child only if its state is not in reached and never replace a state; UCS and A* also add it when it reaches a known state with a strictly lower g (replace).
- **Stale-node rule** (UCS and A*): a removed node whose g exceeds the g stored in reached is discarded silently and is not counted as expanded. GBFS never replaces, so it never has stale nodes.
- **New tools:** `check_heuristic`, `--batch`, `--scan`, `--trace`, `--verbose`, `--no-reached`; the algorithm prompt accepts BFS, DFS, UCS, IDS, GBFS, A* and ALL (case-insensitive, spaces ignored). BFS, DFS, UCS and IDS never read h or COORDINATES; graph, coordinates and goal are passed to every search as arguments.

2.2 The one generic loop, and where each strategy plugs in

1. Make root, put on FRONTIER *plug-in: frontier type*	2. POP a node *plug-in: key fn*	3. STALE? g > reached[g] → discard *plug-in: reopens flag*	4. GOAL TEST (on removal) → return	5. EXPAND count + 1, children A–Z	6. REACHED RULE keep / replace / drop each child *plug-in: reopens flag*	7. ADD kept children to FRONTIER *plug-in: key fn*

Read left to right; after step 7 the loop returns to step 2. An empty frontier means "No route found."; the 10,000-expansion cap is checked just before step 5. IDS reuses the same loop with a depth limit (a node at the limit is goal-tested but not expanded) and a path cycle check instead of reached.

```
search(graph, source, goal, strategy, coordinates, use_reached, trace):
    frontier <- strategy.make_frontier(goal, coordinates)   # FIFO | LIFO | heap(key_fn)
    frontier.add([root]); reached <- {source: root}         # reached = None if tree-like
    while frontier is not empty:
        node <- frontier.pop()                              # heap order: (key, name, counter)
        if node.g > reached[node.state].g (UCS, A* only):   # STALE: cheaper path found later
            continue                                        #   discard, do not count
        if node.state == goal: return node                  # goal test on REMOVAL
        if expanded >= 10000:    return CAP_REACHED
        expanded <- expanded + 1
        for child in expand(node):                          # neighbours in A-Z order
            seen <- reached.get(child.state)
            if seen exists and not (strategy.reopens and child.g < seen.g):
                continue                                    # REACHED RULE: drop the child
            reached[child.state] <- child                   # record, or replace the old node
            kept.append(child)
        frontier.add(kept)                                  # DFS pushes them reversed
    return FAILURE ("No route found.")
```

Strategy	Frontier	Key (smaller first)	Reached rule: add child when…
BFS	FIFO queue	–	state not in reached
DFS	LIFO stack (children pushed reversed)	–	state not in reached
UCS	heap	g	not in reached, or lower g than stored → replace; stale nodes dropped
GBFS	heap	h	not in reached; never replaced; never reads g
A*	heap	g + h	not in reached, or lower g → replace; stale nodes dropped
IDS	the same loop with a LIFO stack, run by a driver	–	state not on the path from the root; limits 0, 1, 2, … until the goal is found; expansions summed over all iterations

2.3 Program flow from the keyboard to the answer

- `main()` parses the flags. `--check-heuristic`, `--batch` and `--scan` run their tool and exit. Otherwise it validates source, destination and algorithm (case-insensitive, trimmed); an unknown place or algorithm prints a message (and the list of valid places) and asks again. If source equals destination, every algorithm prints a one-place route with cost 0, hops 0 and 0 nodes expanded.
- For each requested algorithm (ALL = BFS, DFS, UCS, IDS, GBFS, A*, one block each, blank line between), `run_algorithm` either calls the generic search directly or calls the IDS driver, which reruns the generic search with a depth limit and a path cycle check. The only "which algorithm?" decision is made there, outside the loop.
- `format_result` prints Algorithm / Route / Cost (metres) / Hops / Nodes expanded; GBFS and A* add Heuristic at source; IDS adds its final depth limit; `--verbose` adds Max frontier size and per-limit IDS lines. Cost is recomputed from the route by `route_cost`, so it is independent of the search.
- Tree-like mode (`--no-reached`): no reached table, nothing is discarded, cycles are not detected; the cap of 10,000 expansions ends the search with "Search stopped: expansion cap (10000) reached."
- `--trace` prints push / pop / stale / replace events. `--batch` runs the ten required problems × six algorithms and writes the 60-row `results_log.csv`. `--scan` runs UCS, A* and GBFS on all 182 ordered school pairs and prints the summary, counting a "reopening" whenever a child reaches an already-expanded place with a lower g.
- For example (A*, Social Sciences → Statistics and Planning, `--trace`, Appendix B3): A* first reaches Statistics via Central Teaching Facility at g = 840 (push, f = 840). Later it finds Main Library at g = 300 instead of 350 (replace), and CoCIS's cheaper link gives Statistics g = 770 (replace 840 → 770); the g = 770 node is popped as the goal: cost 770, 7 expansions.

3. Verification (B4)

Check	Required	Obtained	Result
Consistency	864 checks, 0 violations	864 checks, 0 violations	Pass
Admissibility	306 pairs, 0 violations	306 pairs, 0 violations	Pass
Law → Education (h = 311)	UCS 430/3/6, A* 430/3/3, GBFS 430/3/3	UCS 430/3/6, A* 430/3/3, GBFS 430/3/3 (h = 311)	Pass
CoCIS → Agricultural Sciences (h = 1274)	UCS 1480/7/16, A* 1480/7/10, GBFS 1480/7/7	UCS 1480/7/16, A* 1480/7/10, GBFS 1480/7/7 (h = 1274)	Pass
Social Sciences → Statistics (h = 605)	UCS 770/5/11, A* 770/5/7, GBFS 840/4/4	UCS 770/5/11, A* 770/5/7, GBFS 840/4/4 (h = 605)	Pass
Tree-like: Margaret Trowell → Vet. Medicine	UCS hits cap; A* 2130 after 112; GBFS 2280 after 8	UCS cap (10000); A* 2130 / 112; GBFS 2280 / 8	Pass
Scan totals	182, 161, 21, 63, 99, 53, 1801 / 1008 / 731, 0, 124, 250	182, 161, 21, 63, 99, 53, 1801 / 1008 / 731, 0, 124, 250	Pass

Why A* succeeds in tree-like mode where UCS gives up. Without a reached table, UCS must expand every walk whose g is below the optimal 2130 m. Short edges (70, 90, 100 m) can be walked back and forth, so the number of such walks explodes and passes 10,000 long before the goal is reached. A* only expands nodes with f = g + h ≤ 2130. Walking back and forth adds at least twice an edge weight to g while h returns to the same value, so looping nodes quickly exceed f = 2130 and are never popped; the goal (f = 2130) is removed after 112 expansions. Neither cycles nor the cap stop A*: the cycles are priced out by f, and 112 is far below 10,000. GBFS also finishes (8 expansions) but returns 2280 m, not the optimum.

Reopening. The scan reports 0 reopened states because h is consistent: the first removal of a state already has its cheapest g. If it were not 0 (an admissible but inconsistent h), A* would still be correct: the lower-g rule lets a cheaper path replace the old node, the state is expanded again, and the old entry is dropped as stale.

4. Results (Part C)

4.1 Table 1 – cost in metres / nodes expanded (ten required problems)

#	BFS	DFS	UCS	IDS	GBFS	A*
1	530 / 10	530 / 15	530 / 10	530 / 8	530 / 3	530 / 4
2	1140 / 13	1140 / 4	1140 / 17	1140 / 18	1140 / 4	1140 / 5
3	**1170 / 12**	**1170 / 9**	870 / 11	**1170 / 32**	870 / 5	870 / 7
4	**2250 / 17**	**2340 / 11**	2130 / 17	**2250 / 324**	**2280 / 8**	2130 / 17
5	**900 / 9**	**1680 / 7**	770 / 11	**900 / 9**	**900 / 3**	770 / 4
6	100 / 3	100 / 17	100 / 1	100 / 1	100 / 1	100 / 1
7	**860 / 8**	**860 / 13**	850 / 11	**860 / 7**	**860 / 3**	850 / 5
8	**1140 / 14**	920 / 5	920 / 13	**1140 / 24**	**1140 / 4**	920 / 7
9	1110 / 11	1110 / 12	1110 / 9	1110 / 21	**1360 / 5**	1110 / 5
10	**1990 / 15**	1870 / 9	1870 / 17	**1990 / 43**	**1990 / 7**	1870 / 11

Bold cells: cost above the UCS (optimal) cost.

4.2 Table 2 – per algorithm (optimal cost = UCS cost)

Algorithm	Problems solved at the optimal cost (of 10)	Total nodes expanded (ten problems)	Most nodes on one problem
BFS	4	112	17
DFS	6	102	17
UCS	10	117	17
IDS	4	487	324
GBFS	4	43	8
A*	10	66	17

4.3 C2 – greedy beats A* on nodes but loses on cost

GBFS expands fewer nodes but returns a costlier route on problems 4, 5, 7, 8, 10 (5 of ten). Two cases in detail (g, h, f = g + h at the first split, read from `--trace`):

Case	A* route (cost m / nodes)	GBFS route (cost m / nodes)	First split and competing nodes
Problem 4 Margaret Trowell → Veterinary Medicine	Margaret Trowell → Engineering → CoCIS → Central Teaching Facility → Freedom Square → Main Library → Education → Natural Sciences → Food Technology → Agricultural Sciences → Veterinary Medicine **2130 m / 17**	Margaret Trowell → Liberal and Performing Arts → Main Building → Social Sciences → Education → Natural Sciences → Food Technology → Agricultural Sciences → Veterinary Medicine **2280 m / 8 (+150 m)**	Split at the **source**. Engineering: g 140, h 1665, f 1805. Liberal & Perf. Arts: g 450, h 1371, f 1821. GBFS pops Liberal Arts (h 1371 < 1665); A* pops Engineering (f 1805 < 1821).
Scan pair (largest extra among the 53) Natural Sciences → School of the Built Environment	Natural Sciences → Education → Main Library → Freedom Square → Central Teaching Facility → CoCIS → Engineering → Built Environment **1240 m / 9**	Natural Sciences → Education → Main Library → Main Building → Liberal and Performing Arts → Margaret Trowell → Built Environment **1480 m / 6 (+240 m)**	Split at **Main Library**. Freedom Square: g 550, h 570, f 1120. Main Building: g 610, h 569, f 1179. GBFS: 569 < 570 → Main Building. A*: 1120 < 1179 → Freedom Square.

Why this happens. GBFS looks only at h, so a 1 m advantage (569 vs 570) or a 294 m advantage in straight-line distance outweighs the 60–310 m of extra walking already paid, and it never revisits the choice. A* adds g, sees the extra walking, and prefers the cheaper branch. (Four other scan pairs share the 240 m figure; the 250 m maximum of the whole scan occurs on pairs where GBFS and A* expand the same number of nodes, so they are not in this list.)

Problem 9 is a decoy. GBFS returns 1360 m against A*'s 1110 m, but both expand exactly 5 nodes (UCS: 9), so GBFS did not win on nodes. Both expand the same first four places (Food Technology, Natural Sciences, Education, Main Library). For the fifth, GBFS pops School of Economics (h 106 < Freedom Square's 174) and A* pops Freedom Square (f 1104 < 1346); each then pops the goal, School of Law, next. Same count, different route.

4.4 C3 – strength of the heuristic (scan, H_SCALE_PERCENT varied, then reset to 100)

H_SCALE_PERCENT	Pairs where A* returned the UCS cost (of 182)	Total nodes expanded by A*	A* reopened states
0	182	1801	0
50	182	1443	0
100	182	1008	0
150	162	799	1
200	142	760	0

At scale 0 we got 182 / 1801 / 0, which is identical to UCS because f = g, so A* uses UCS's key, tie-break and reached rule and expands exactly the same nodes.

5. Analysis

1. **A* is optimal here.** Graph-search A* is optimal when (a) h is admissible (consistent is the stronger condition needed with a reached table), and (b) the goal test is applied when a node is removed, not generated. (a) holds because every edge weight ≥ straight-line distance (A2, A3); `--check-heuristic` found 0 violations in 864 consistency checks and 306 admissibility pairs. (b) is built into the loop (step 4 of Section 2.2). Evidence: A* returned the UCS cost on all 182 scanned pairs and on all ten problems (Table 2: 10/10). When we break (a) at scales 150 and 200, A* drops to 162 and 142 of 182.
2. **Why greedy is fast but unreliable.** GBFS gains speed: 43 vs 66 (A*) vs 117 (UCS) total nodes on the ten problems, and 731 vs 1008 vs 1801 over the scan; it expanded fewer nodes than A* on 99 of 182 pairs. It loses reliability: only 4 of 10 problems optimal, and costlier than A* on 63 of 182 pairs (34.6 %), by 124 m on average and up to 250 m. The C2 cases show the mechanism: it ignores g.
3. **Where the heuristic misleads.** In the Natural Sciences → Built Environment case, the Built Environment is only 569 m (straight line) from Main Building, but the walk goes through Liberal and Performing Arts by the 350 m and 450 m edges. From Main Library the true remaining cost via Main Building is 1020 m (h = 569, a gap of 451 m) while via

Freedom Square it is 780 m (h = 570, a gap of 210 m). A 1 m difference in h hid a 240 m difference in true cost. Likewise Central Teaching Facility → Statistics is 300 m on foot but only 180 m on the map. The larger the gap between h and the true remaining cost, and the more unevenly the gap is spread across branches, the more a heuristic can mislead.

4. **A* versus UCS.** Over the ten problems A* expanded 66 nodes against UCS's 117 (44 % fewer); over the scan 1008 against 1801 (also 44 % fewer; 161 pairs fewer, 21 the same). On problem 4 A* expands 17 = UCS's 17: the graph has 18 places, so both expanded every place except the goal and there was nothing left to prune. h favours the Liberal Arts / Main Building side (h 1371 and 1152) while the optimal route runs through Engineering and CoCIS, so A* has to examine both sides before it can be sure.
5. **Strength of the heuristic.** From scale 0 to 200, A*'s nodes fall steadily (1801 → 1443 → 1008 → 799 → 760, approaching GBFS's 731) but optimality holds only up to 100: 0 gives UCS; 50 and 100 keep A* optimal (still admissible and consistent: scaling by ≤ 1 only lowers h); 150 and 200 violate admissibility (h overestimates), so A* returns costlier routes on 20 and 40 pairs. Reopenings were 0, 0, 0, 1, 0: at 150 h is no longer consistent, so one place was expanded before its cheaper path was found; at 200 the search is nearly greedy and commits early, so its errors show up as wrong answers rather than as reopenings.
6. **Memory** (problem 4, `--verbose`, max frontier size). IDS 12 > DFS 6 = A* 6 > BFS 5 = GBFS 5 > UCS 4. All are small because the campus is a sparse, chain-like graph. IDS is largest because it has no reached table (only a path check), so siblings and duplicates of the same place pile up on its stack during its 8-hop pass; the others use a reached table that suppresses duplicates. On a large graph the textbook ranking (DFS/IDS use the least memory, BFS the most) would dominate; here the presence of a reached table matters more.
7. **Choosing a strategy.** For a phone app answering thousands of requests per second we would choose A*: always optimal here (182/182), 44 % fewer expansions than UCS, frontier ≤ 6 entries. The trade-off is one square root per generated node and the need to ship map coordinates. GBFS is 27 % cheaper again (731 vs 1008) but gives a costlier route for one request in three, about 124 m too long. With only 306 ordered pairs, a real app could also precompute all routes and simply look them up.
8. **A better heuristic.** One idea is to use landmark (ALT) distances: pick a few places (e.g. Freedom Square, Main Library), run UCS once from each to get exact distances d(L, x), and use h(n) = max over L of |d(L, goal) − d(L, n)|, also taking the maximum with the straight-line value. By the triangle inequality this never overestimates, and it "knows" about long winding edges such as CTF → Statistics (300 m). Cost: k UCS runs of preprocessing and a table of k × 18 numbers; each h evaluation costs O(k) instead of one square root.

6. Reflection

What surprised us most is how small the difference was when GBFS made its wrong choice: at Main Library the two candidate places differed by only 1 m of straight-line distance (569 vs 570), yet by 240 m of real walking. We also did not expect plain DFS to find the optimal route on 6 of 10 problems (more than BFS or IDS), or A* to be no better than UCS on problem 4. We learned that a heuristic only helps as much as the shape of the graph lets it.

Appendix A – results_log.csv (60 data rows, produced by --batch)

Columns: problem, source, destination, algorithm, route, cost_m, hops, nodes_expanded. Source and destination are shown once per problem.

Alg.	Route	Cost m	Hops	Nodes
Problem 1: CoCIS → School of Law				
BFS	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	10
DFS	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	15
UCS	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	10
IDS	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	8
GBFS	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	3
A*	CoCIS → Central Teaching Facility → Freedom Square → School of Law	530	3	4
Problem 2: School of Education → School of Veterinary Medicine				
BFS	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	13
DFS	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	4
UCS	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	17
IDS	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	18
GBFS	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	4
A*	School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	1140	4	5
Problem 3: School of the Built Environment → School of Law				
BFS	School of the Built Environment → Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → Freedom Square → School of Law	1170	5	12
DFS	School of the Built Environment → Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → Freedom Square → School of Law	1170	5	9
UCS	School of the Built Environment → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → School of Law	870	5	11
IDS	School of the Built Environment → Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → Freedom Square → School of Law	1170	5	32
GBFS	School of the Built Environment → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → School of Law	870	5	5
A*	School of the Built Environment → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → School of Law	870	5	7
Problem 4: Margaret Trowell School of Industrial and Fine Art → School of Veterinary Medicine				
BFS	Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → Main Library → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2250	8	17
DFS	Margaret Trowell School of Industrial and Fine Art → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → Main Building → School of Social Sciences → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2340	11	11
UCS	Margaret Trowell School of Industrial and Fine Art → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → Main Library → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2130	10	17
IDS	Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → Main Library → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2250	8	324
GBFS	Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts → Main Building → School of Social Sciences → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2280	8	8
A*	Margaret Trowell School of Industrial and Fine Art → School of Engineering → CoCIS → Central Teaching Facility → Freedom Square → Main Library → School of Education → College of Natural Sciences → School of Food Technology, Nutrition and Bioengineering → School of Agricultural Sciences → School of Veterinary Medicine	2130	10	17
Problem 5: School of Economics → School of Liberal and Performing Arts				
BFS	School of Economics → Main Library → Main Building → School of Liberal and Performing Arts	900	3	9
DFS	School of Economics → Main Library → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering → Margaret Trowell School of Industrial and Fine Art → School of Liberal and Performing Arts	1680	7	7
UCS	School of Economics → School of Law → Freedom Square → Main Building → School of Liberal and Performing Arts	770	4	11
IDS	School of Economics → Main Library → Main Building → School of Liberal and Performing Arts	900	3	9
GBFS	School of Economics → Main Library → Main Building → School of Liberal and Performing Arts	900	3	3

A*	School of Economics → School of Law → Freedom Square → Main Building → School of Liberal and Performing Arts	770	4	4
Problem 6: CoCIS → School of Statistics and Planning				
BFS	CoCIS → School of Statistics and Planning	100	1	3
DFS	CoCIS → School of Statistics and Planning	100	1	17
UCS	CoCIS → School of Statistics and Planning	100	1	1
IDS	CoCIS → School of Statistics and Planning	100	1	1
GBFS	CoCIS → School of Statistics and Planning	100	1	1
A*	CoCIS → School of Statistics and Planning	100	1	1
Problem 7: College of Natural Sciences → School of Economics				
BFS	College of Natural Sciences → School of Education → Main Library → School of Economics	860	3	8
DFS	College of Natural Sciences → School of Education → Main Library → School of Economics	860	3	13
UCS	College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law → School of Economics	850	5	11
IDS	College of Natural Sciences → School of Education → Main Library → School of Economics	860	3	7
GBFS	College of Natural Sciences → School of Education → Main Library → School of Economics	860	3	3
A*	College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law → School of Economics	850	5	5
Problem 8: School of Social Sciences → School of Engineering				
BFS	School of Social Sciences → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art → School of Engineering	1140	4	14
DFS	School of Social Sciences → Main Building → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering	920	5	5
UCS	School of Social Sciences → Main Building → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering	920	5	13
IDS	School of Social Sciences → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art → School of Engineering	1140	4	24
GBFS	School of Social Sciences → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art → School of Engineering	1140	4	4
A*	School of Social Sciences → Main Building → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering	920	5	7
Problem 9: School of Food Technology, Nutrition and Bioengineering → School of Law				
BFS	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law	1110	5	11
DFS	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law	1110	5	12
UCS	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law	1110	5	9
IDS	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law	1110	5	21
GBFS	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → School of Economics → School of Law	1360	5	5
A*	School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → School of Law	1110	5	5
Problem 10: School of Agricultural Sciences → Margaret Trowell School of Industrial and Fine Art				
BFS	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art	1990	7	15
DFS	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering → Margaret Trowell School of Industrial and Fine Art	1870	9	9
UCS	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering → Margaret Trowell School of Industrial and Fine Art	1870	9	17
IDS	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art	1990	7	43
GBFS	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Main Building → School of Liberal and Performing Arts → Margaret Trowell School of Industrial and Fine Art	1990	7	7
A*	School of Agricultural Sciences → School of Food Technology, Nutrition and Bioengineering → College of Natural Sciences → School of Education → Main Library → Freedom Square → Central Teaching Facility → CoCIS → School of Engineering → Margaret Trowell School of Industrial and Fine Art	1870	9	11

Appendix B – pasted --trace output

B1. GBFS, problem 4 (complete trace)

```
push Margaret Trowell School of Industrial and Fine Art g=0 h=1723 key=1723
pop Margaret Trowell School of Industrial and Fine Art key=1723
push School of Engineering g=140 h=1665 key=1665
push School of Liberal and Performing Arts g=450 h=1371 key=1371
push School of the Built Environment g=70 h=1721 key=1721
pop School of Liberal and Performing Arts key=1371
push Main Building g=800 h=1152 key=1152
pop Main Building key=1152
push Freedom Square g=920 h=1172 key=1172
push Main Library g=950 h=1093 key=1093
push School of Social Sciences g=1000 h=989 key=989
pop School of Social Sciences key=989
push School of Education g=1140 h=955 key=955
pop School of Education key=955
push College of Natural Sciences g=1440 h=695 key=695
pop College of Natural Sciences key=695
push School of Food Technology, Nutrition and Bioengineering g=1820 h=375 key=375
pop School of Food Technology, Nutrition and Bioengineering key=375
push School of Agricultural Sciences g=2020 h=212 key=212
pop School of Agricultural Sciences key=212
push School of Veterinary Medicine g=2280 h=0 key=0
pop School of Veterinary Medicine key=0
```

B2. A*, problem 4 (first 24 events – the split at the source is visible in the first push lines)

```
push Margaret Trowell School of Industrial and Fine Art g=0 h=1723 key=1723
pop Margaret Trowell School of Industrial and Fine Art key=1723
push School of Engineering g=140 h=1665 key=1805
push School of Liberal and Performing Arts g=450 h=1371 key=1821
push School of the Built Environment g=70 h=1721 key=1791
pop School of the Built Environment key=1791
pop School of Engineering key=1805
push CoCIS g=390 h=1486 key=1876
pop School of Liberal and Performing Arts key=1821
push Main Building g=800 h=1152 key=1952
pop CoCIS key=1876
push Central Teaching Facility g=520 h=1377 key=1897
push School of Statistics and Planning g=490 h=1519 key=2009
pop Central Teaching Facility key=1897
push Freedom Square g=740 h=1172 key=1912
pop Freedom Square key=1912
push Main Library g=830 h=1093 key=1923
push School of Law g=920 h=1184 key=2104
pop Main Library key=1923
push School of Economics g=1230 h=1265 key=2495
push School of Education g=990 h=955 key=1945
pop School of Education key=1945
push College of Natural Sciences g=1290 h=695 key=1985
push School of Social Sciences g=1130 h=989 key=2119
```

B3. A*, Social Sciences → Statistics and Planning (shows push and replace; worked example in Section 2.3)

```
push School of Social Sciences g=0 h=605 key=605
pop School of Social Sciences key=605
push Main Building g=200 h=421 key=621
push School of Education g=140 h=581 key=721
pop Main Building key=621
push Freedom Square g=320 h=358 key=678
push Main Library g=350 h=436 key=786
push School of Liberal and Performing Arts g=550 h=545 key=1095
pop Freedom Square key=678
push Central Teaching Facility g=540 h=180 key=720
push School of Law g=500 h=352 key=852
pop Central Teaching Facility key=720
push CoCIS g=670 h=87 key=757
push School of Statistics and Planning g=840 h=0 key=840
pop School of Education key=721
replace Main Library g=350 -> 300
push College of Natural Sciences g=440 h=828 key=1268
push Main Library g=300 h=436 key=736
pop Main Library key=736
push School of Economics g=700 h=316 key=1016
pop CoCIS key=757
replace School of Statistics and Planning g=840 -> 770
push School of Engineering g=920 h=272 key=1192
push School of Statistics and Planning g=770 h=0 key=770
pop School of Statistics and Planning key=770
```